

AVERAGES

Problems

- 1) The average of five positive numbers is 213. The average of the first two numbers is 233.5 and the average of last two numbers is 271. What is the third number?
a) 64 b) 56 c) 106 d) Cannot be determined e) None of these
- 2) Find the average of the following set of scores:
568, 460, 349, 987, 105, 178, 426
a) 442 b) 441 c) 440 d) 439 e) None of these
- 3) If $36a + 36b = 576$, then what is the average of a and b?
a) 16 b) 8 c) 12 d) 6 e) None of these
- 4) The average marks of 65 students in a class was calculated as 150. It was later realized that the marks of one of the students was calculated as 142, whereas his actual average marks were 152. What is the actual average marks of the group of 65 students? Rounded off to two digits after decimal)
a) 151.25 b) 150.15 c) 151.10 d) 150.19 e) None of these
- 5) In a class there are 32 boys and 28 girls. The average age of the boys in the class is 14 years and the average age of the girls in the class is 13 years. What is the average age of the whole class? Rounded off to two digits after decimal)
a) 13.50 b) 13.53 c) 12.51 d) 13.42 e) None of these
- 6) The average of 5 consecutive even numbers A, B, C, D and E respectively is 74. What is the product of C and E?
a) 5928 b) 5616 c) 5538 d) 5772 e) None of these
- 7) Average of four consecutive odd numbers is 106. What is the third number in ascending order?
a) 107 b) 111 c) 113 d) Cannot be determined e) None of these
- 8) If average of 11 consecutive odd numbers is 17, what is the difference between the smallest and the largest number?
a) 18 b) 20 c) 22 d) 24 e) None of these
- 9) Of the three numbers, the average of the first and the second is greater than the average of the second and the third by 15. What is the difference between the first and the third of the three numbers?
a) 15 b) 45 c) 60 d) Data inadequate e) None of these
- 10) The ratio of roses and lilies in a garden is 3:2 respectively. The average number of roses and lilies is 180. What is the number of lilies in the garden?
a) 144 b) 182 c) 216 d) 360 e) None of these
- 11) The average monthly income of a family of four earning members was Rs.15130. One of the daughters in the family got married and left home, so the average monthly income of the family came down to Rs.14660. What is the monthly income of the married daughter?
a) Rs.15350 b) Rs.12000 c) Rs.16540 d) Cannot be determined e) None of these

12) The average temperature of Monday, Tuesday, Wednesday and Thursday was 36.5°C and for Tuesday, Wednesday, Thursday and Friday was 34.5°C . If the temperature on Monday was 38°C , find the temperature on Friday.

a) 34°C b) 36°C c) 37.4°C d) 32°C e) 30°C

13) A cricketer has an average score of 49 runs in 24 innings. How many runs must he score in the 25th innings to make his average 50?

a) 94 b) 84 c) 74 d) 76 e) None of these

14) The average age of 14 boys in a class is 13 years. If the class teacher's age is included, the average age is increased by one year. What is the class teacher's age?

a) 31 years b) 27.5 years c) 24 years d) 28 years e) None of these

15) The average of four positive integers is 73.5. The highest integer is 108 and the lowest integer is 29. The difference between the remaining two integers is 15. Which of the following is the smaller of the remaining two integers?

a) 80 b) 86 c) 73 d) Cannot be determined e) None of these

16) In a class of 75 students, the average age is 23 years. The average age of male students is 25 years and that of female students is 20 years. Then the ratio of male to female students is

a) 3:2 b) 7:3 c) 8:7 d) 6:9 e) 9:6

17) Average salary of 19 workers in an industry Rs.2500. The salary of supervisor is Rs.5550. Find the average salary of all the 20 employees.

a) Rs.3355.5 b) Rs.4500.00 c) Rs.4642.5 d) Rs.2652.5 e) None of these

18) The average of three even consecutive numbers is 24. What is the summation of the three numbers?

a) 24 b) 72 c) 26 d) 80 e) None of these

19) The sum of seven consecutive even numbers of a set is 532. What is the average of first four consecutive even numbers of the same set?

a) 76 b) 75 c) 74 d) 73 e) None of these

20) There are 50 boys in a class. One boy weighing 40 kg goes away and at the same time another boy joins the class. If the average weight of the class is thus decreased by 100 g, find the weight of the new boy.

a) 35kg b) 43kg c) 36kg d) 30 kg e) None of these

21) Kamlesh bought 65 books for Rs.1050 from one shop and 50 books for Rs.1020 from another. What is the average price he paid per book?

a) Rs. 36.40 b) Rs.18.20 c) Rs. 24 d) Rs.18 e) None of these

22) The average weight of 12 boys is 35 kgs. If the weight of an adult is added, the average becomes 37 kgm. What is the weight of the adult?

a) 65kgs b) 68 kgs c) 62 kgs d) 63 kgs e) None of these

23) The average marks in Science subject of a class of 20 students is 68. If the marks of two students were misread as 48 and 65 of the actual marks 72 and 61 respectively, then what would be the correct average?

- a) 68.5 b) 69 c) 69.5 d) 70 e) 66

24) The average marks of nine students in a group is 63. Three of them scored 78, 69 and 48 marks. What are the average marks of remaining six students?

- a) 63.5 b) 64 c) 63 d) 62.5 e) None of these

25) The average age of A, B and C is 26 years. If the average age of A and C is 29 years. What is the age of B?

- a) 26 b) 20 c) 29 d) 23 e) None of these

26) The average of marks obtained in 120 students was 35. If the average of passed candidates was 39 and that of failed candidates is 15, the number of candidates who passed the examination is:

- a) 100 b) 110 c) 120 d) 80

27) The average marks in Hindi subject of a class of 54 students is 76. If the marks of two students are misread as 60 and 77 of the actual marks 36 and 47 respectively, then what would be the correct average?

- a) 75.5 b) 77 c) 75 d) 76.5 e) None of these

AGES

1) 10 years ago, the average age of a family of 4 members was 24 years. Two children having been born with age difference of 2 years, the present average age of the family is the same. The present age of the youngest child is:

- a) 1 year b) 2 years c) 3 years d) 5 years e) None of these

2) Three years ago, the average age of A and B was 18 years. With C joining them, the average age becomes 22 years. How old is C now?

- a) 24 years b) 27 years c) 28 years d) 30 years e) None of these

3) The average age of a husband and wife was 23 years at the time of their marriage. After five years they have a one-year old child. The average age of the family now is:

- a) 19 years b) 23 years c) 28.5 years d) 29.3 years e) None of these

4) A family consists of grandparents, parents and three children. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. What is the average age of the family?

- a) $28\frac{4}{7}$ years b) $31\frac{5}{7}$ years c) $32\frac{1}{7}$ years d) 31 years e) None of these

5) The average age of three boys is 16 years. If their ages are in the ratio 4 : 5 : 7, then the age of youngest boy is :

- a) 8 years b) 9 years c) 12 years d) 16 years e) None of these

6) Average age of 6 sons of a family is 8 years. Average age of the sons together with their parents is 22 years. If the father is older than the mother by 8 years, then the age of the mother is

- a) 44 years b) 52 years c) 60 years d) 68 years e) None of these

7) The average age of the mother and her six children is 12 years which is reduced by 5 years if the mother is excluded. How old is the mother?

- a) 40 years b) 42 years c) 48 years d) 50 years e) None of these

8) The average age of A, B and C is 7 years. The average age of A and B is 6 years and that of B and C is 8 years. Find the age of B.

- a) 6 years b) 7 years c) 8 years d) 9 years e) None of these

9) The average age of 3 friends is 23. Even if the age of the 4th friend is added, the average remains 23. What is the age of the 4th friend?

- a) 21 years b) 23 years c) 32 years d) Cannot be determined e) None of these

10) At present the sum of ages of R and K is 63 years. The ratio of their ages after 7 years will be 7 : 4. What is the present age of R?

- a) 40 years b) 42 years c) 38 years d) 39.5 years e) None of these

11) The ages of K and C are in the ratio 3 : 7. After 2 years their age will be in the ratio 1 : 2. The sum of their ages is

- a) 18 years b) 20 years c) 37 years d) 50 years e) None of these

12) Ratio of A's to P's age is equal to 4 : 3. A will be 26 years old after 6 years. How old is P now?

- a) 12 years b) 21 years c) 15 years d) $19\frac{1}{2}$ years e) None of these

13) 10 years ago A's mother was 4 times older than her daughter. After 10 years the mother will be two times the age of her daughter. The present age of A is

- a) 5 years b) 10 years c) 20 years d) 30 years e) None of these

14) In 10 years, A will be twice as old as B was 10 years ago. If A is now 9 years older than B, the present age of B is

- a) 29 years b) 39 years c) 19 years d) 49 years e) None of these

15) The ratio of the mother's age to the daughter's age is 4 : 1. The product of their ages is 196. The ratio of their ages after 8 years will be

- a) 8 : 3 b) 12 : 5 c) 5 : 3 d) 7 : 3 e) None of these

16) If difference between ages of B and R is 15 years and ratio of their ages is 3 : 2, then the ages of R is

- a) 25 years b) 20 years c) 28 years d) 30 years e) None of these

17) Ten years ago the ratio of ages of P and Q was 5 : 3. If the difference between their ages after 9 years will be 8 years. The present age of P is

- a) 30 years b) 25 years c) 16 years d) 28 years e) None of these

18) The present age difference between father and son is 14 years. The ratio of their ages will be 4 : 3 after 11 years. How old is son now?

- a) 25 years b) 31 years c) 30 years d) 28 years e) None of these

19) The ratio of present ages of x and y is 2 : 3 and the sum of their ages after 4 years will be 68 years. What is the age of y now?

- a) 35 years b) 36 years c) 24 years d) 28 years e) None of these

20) The average age of Rajan and Sajan after a period of 5 years from now will be 20 years. Rajan's present age is 12 years. What is their average age now?

- a) 14 b) 15 c) 10 d) 17 e) None of these

21) The average age of P, Q and R is 5 years more than R's age. If the total ages of P and Q together is 39 years, what is R's age?

- a) 24 years b) 16 years c) 14 years d) 12 years e) None of these

22) The average age of A and B is 54 years, the average age of B and C is 55 years and the average age of C and A is 59 years. What is the age of B in years?

- a) 62 b) 48 c) 56 d) 50 e) None of these

23) Five years ago average age of P and Q was 15 years. Average age of P, Q and R today is 20 years. How old R will be after 10 years?

- a) 10 years b) 20 years c) 30 years d) 40 years e) None of these

24) The age of three members of a family are 5, 10, 15 years respectively. What is their average age?

- a) 10 years b) 5 years c) 15 years d) 20 years e) None of these

25) The average age of a family of five persons is 20 years. If the youngest member is 8 years old. What was the average age of the family at the time of birth of the youngest member?
a) 12 years b) 15 years c) 18 years d) 16 years e) 23 years

26) Sachin is younger than Rahul by 4 years. If their ages are in the respective ratio of 7 : 9, how old is Sachin?
a) 16 years b) 18 years c) 28 years d) Can't be determined e) None of these

27) The ratio between the present ages of P and Q is 6 : 7. If Q is 4 years old than P, what will be the ratio of the ages of P and Q after 4 years?
a) 3 : 4 b) 3 : 5 c) 4 : 3 d) Data is inadequate e) None of these

28) The ratio between the present ages of P and Q is 5 : 7 respectively. If the difference between Q's present age and P's age after 6 years is 2, what is the total of P's and Q's present ages?
a) 48 years b) 52 years c) 56 years d) Cannot be determined e) None of these

29) At present, the ratio between the ages of Arun and Deepak is 4 : 3. After 6 years, Arun's age will be 26 years. What is the age of Deepak at present?
a) 12 years b) 15 years c) $19\frac{1}{2}$ years d) 21 years e) None of these

30) Present ages of X and Y are in the ratio 5 : 6 respectively. Seven years hence this ratio will become 6 : 7 respectively. What is X's present age in years?
a) 35 b) 42 c) $19\frac{1}{2}$ years d) 21 years e) None of these

31) Present ages of Sameer and Anand are in the ratio of 5 : 6 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?
a) 24 b) 27 c) 40 d) Cannot be determined e) None of these

32) Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10. What is Sagar's age at present?
a) 16 years b) 18 years c) 20 years d) Cannot be determined e) None of these

33) The ratio of the present ages of two brothers is 1 : 2 and 5 years back, the ratio was 1 : 3. What will be the ratio of their ages after 5 years ?
a) 1 : 4 b) 2 : 3 c) 3 : 5 d) 5 : 6 e) None of these